

LAB REPORT

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Windows Forensics: Registry Analysis and Event Log Analysis

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For the class of Computer Forensic, taught by

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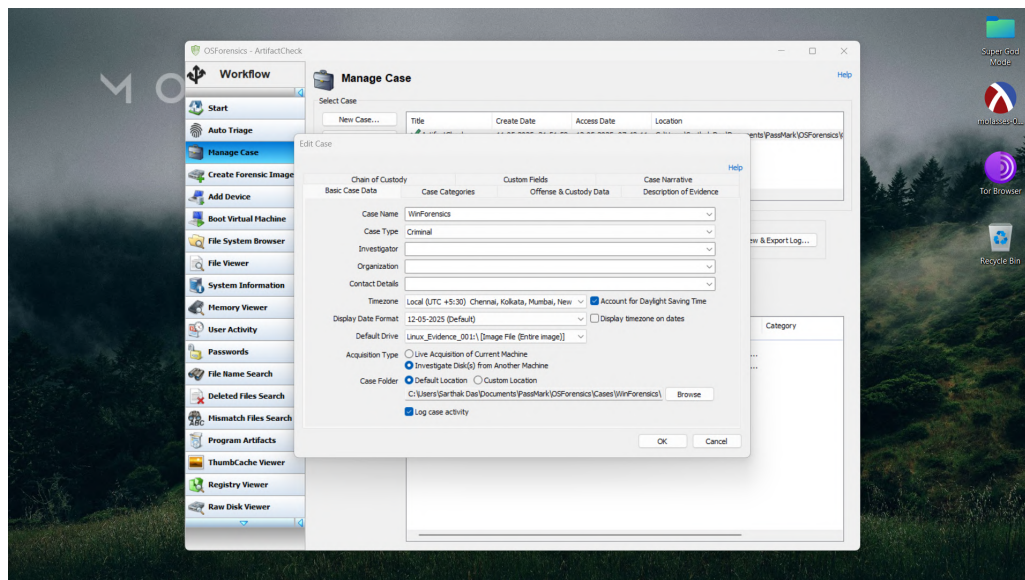
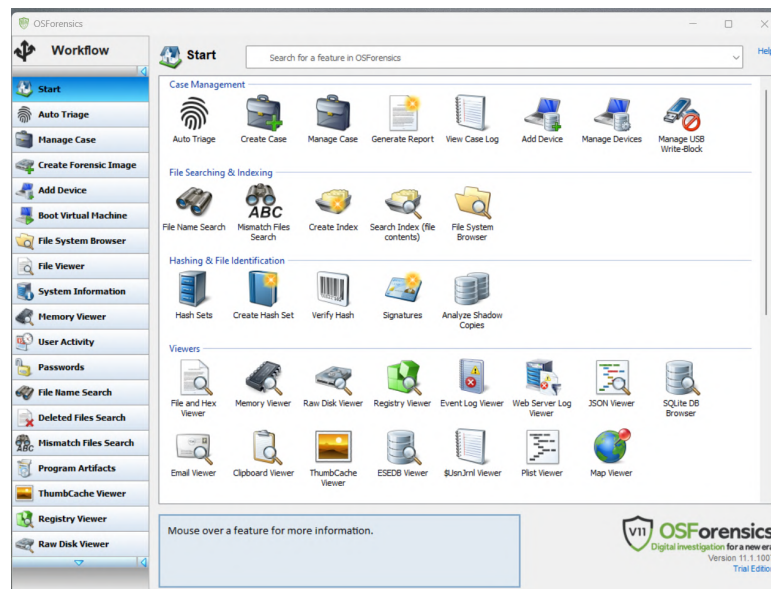
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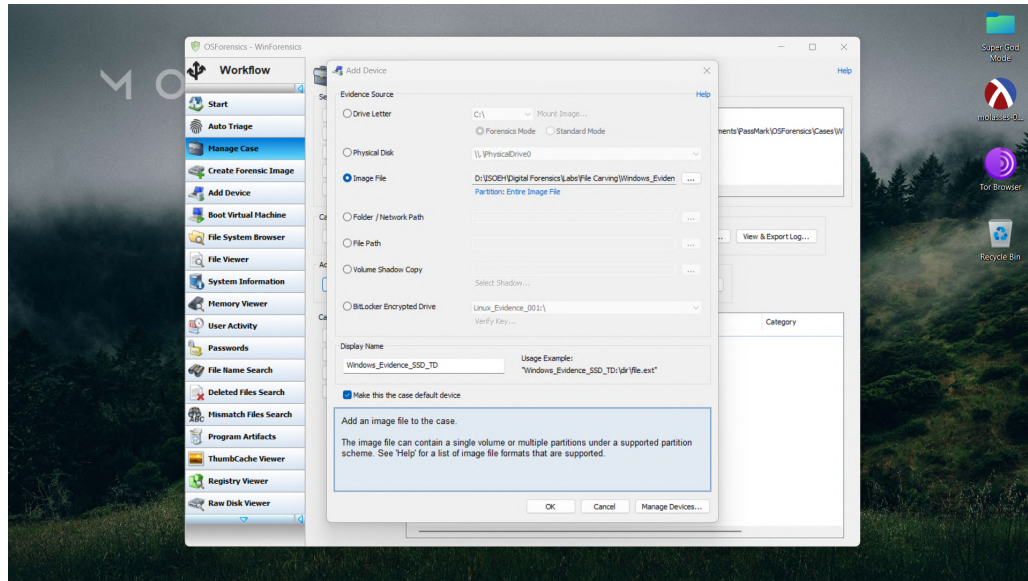
2 Objective 1: Registry Analysis

2.1 Methodology: OSForensics

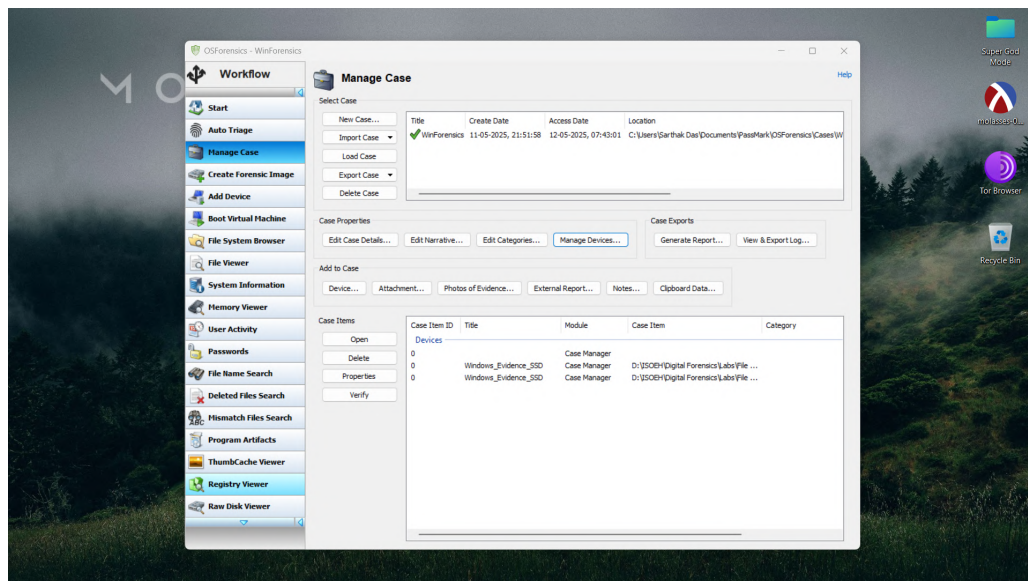
We begin by opening OSForensics, and selecting **Create Case** and set the case name to WinForensics.



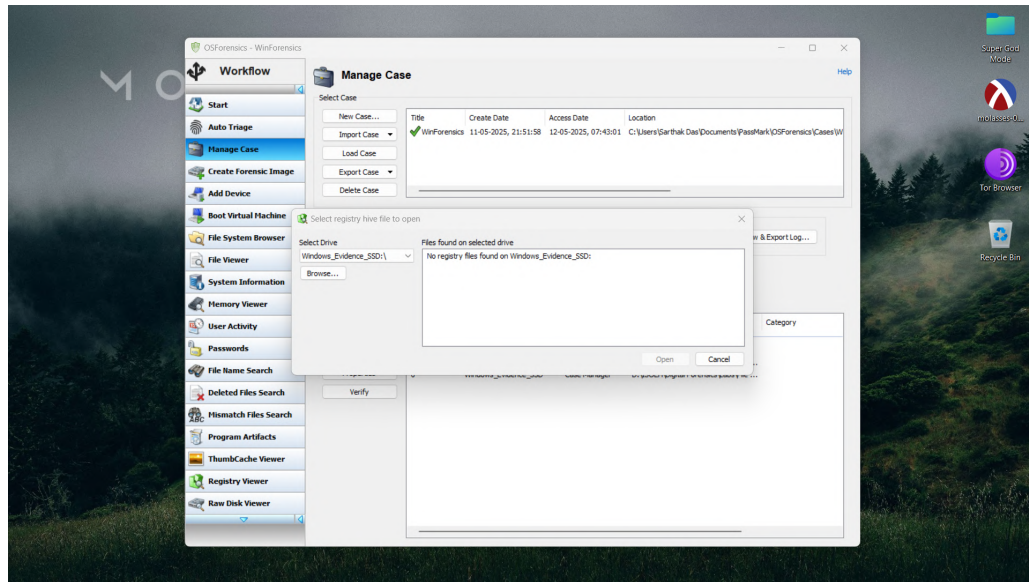
Next, we are prompted to add a device. For the purposes of this lab, we were provided with a Windows disk image, which we add to our case.



Now we can view and analyze the registry in this device by going to **Registry Viewer**.

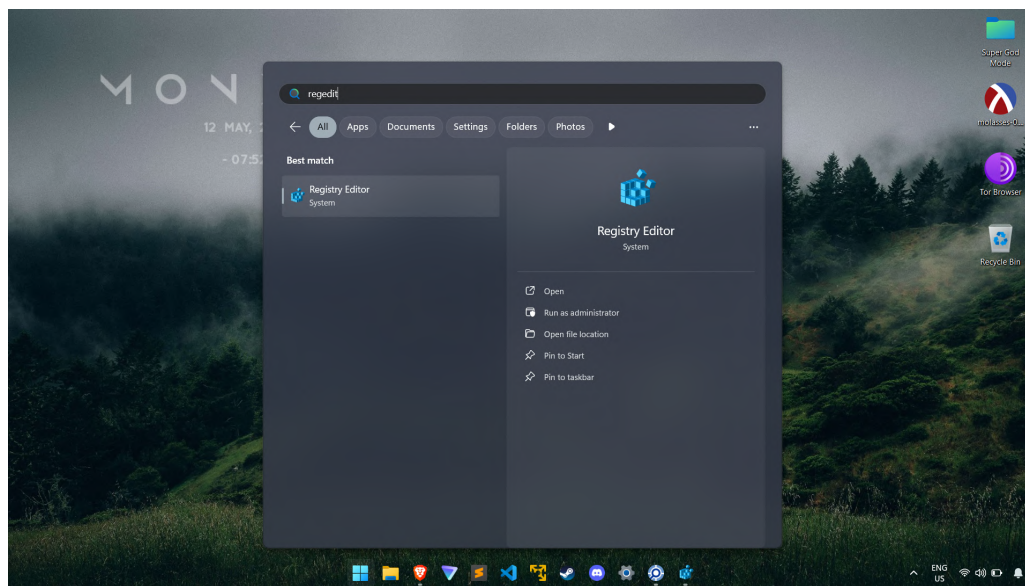


As we can see, no registry entries were found in this disk image.

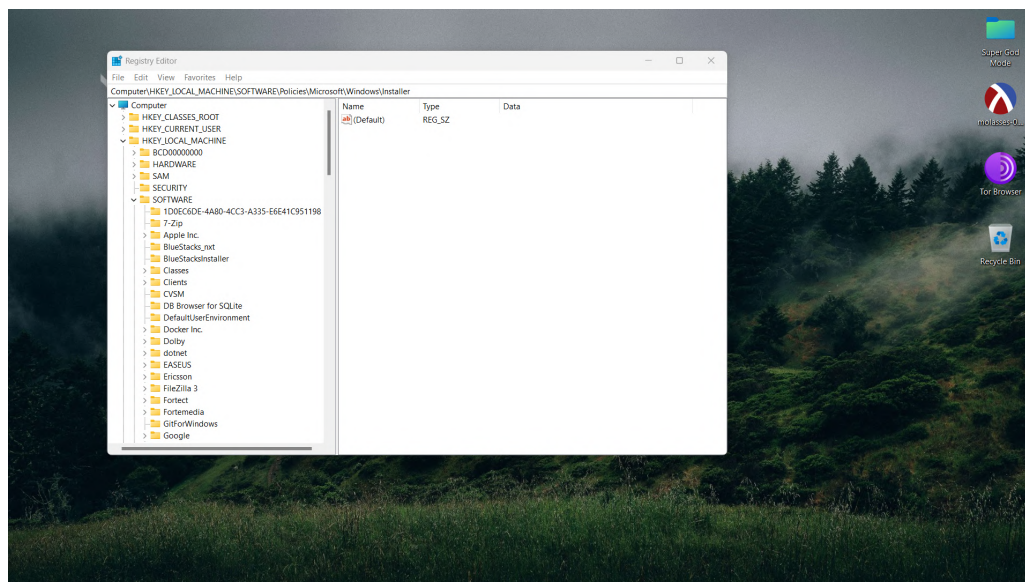


2.2 Methodology: Registry Editor

Windows also provides an inbuilt tool for viewing and editing registries called Registry Editor. Once we live boot from a Windows Disk Image, we can search for *regedit* in the **Start Menu** to open the Registry Editor.



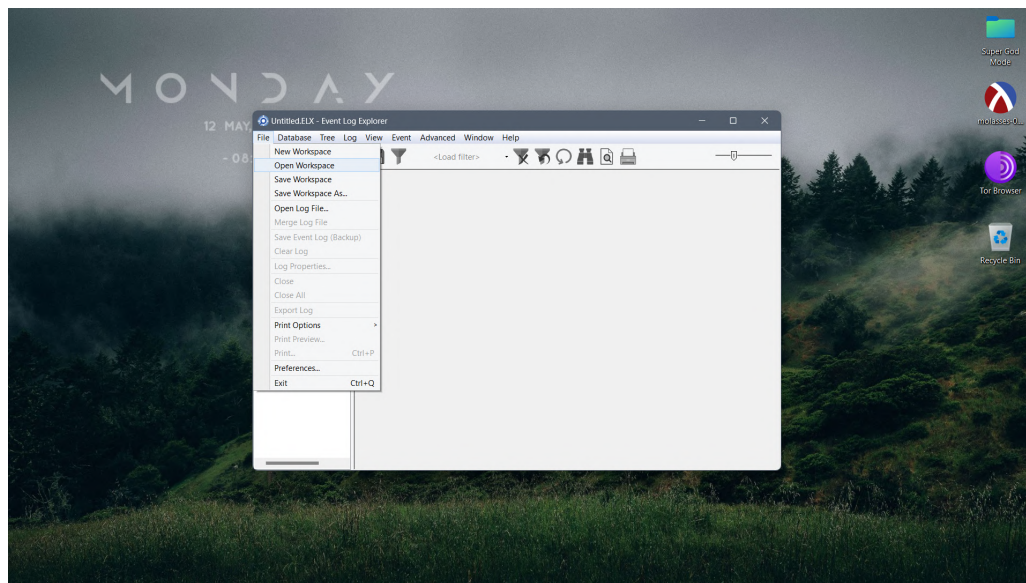
Registry Editor allows us to view, analyze and edit registry entries in our disk image.



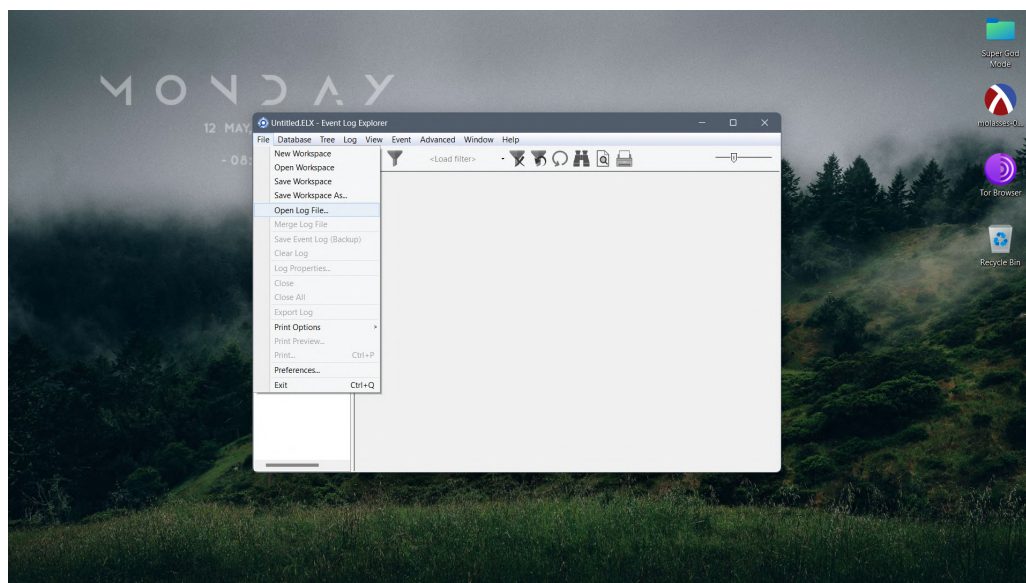
3 Objective 2: Event Log Analysis

3.1 Methodology: Event Log Explorer

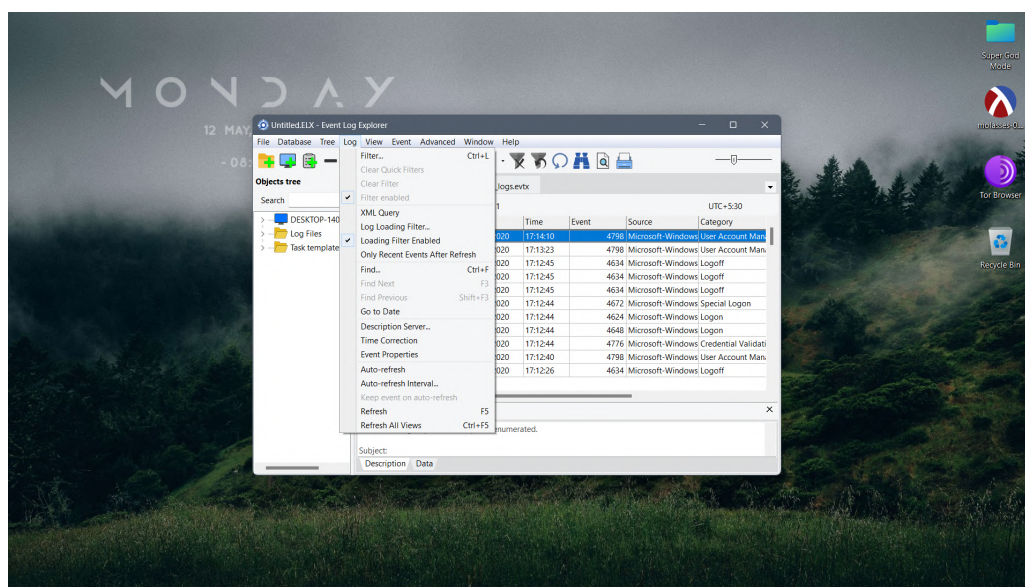
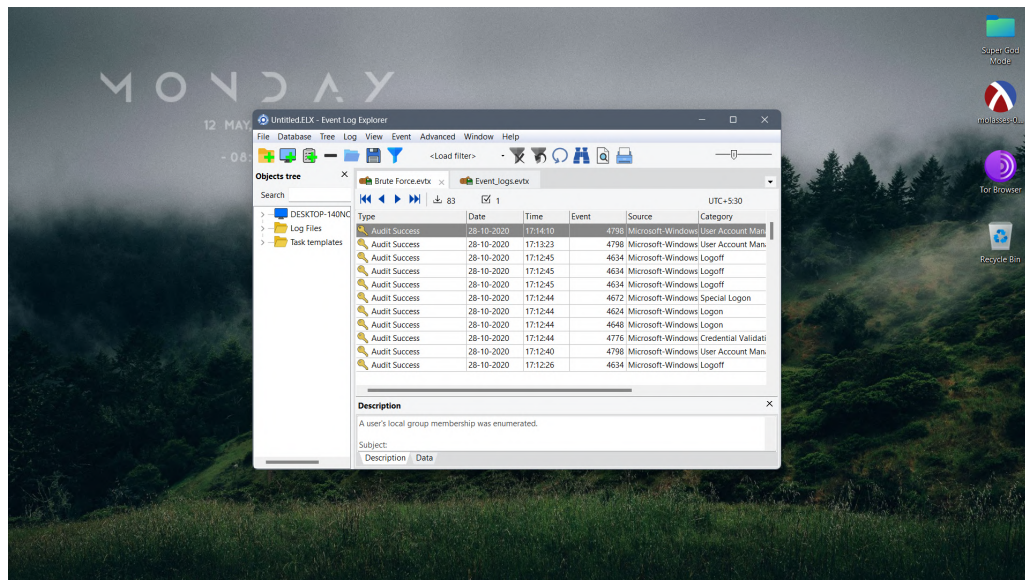
We begin by opening Event Log Explorer, and selecting **File > New Workspace**.



Once the workspace is created we can go to **File > Open Log File** and select the log files we wish to import. For the purposes of this lab, we were provided with two event log files – *Brute Force.evtx* and *Event.logs.evtx*.



Once the log files have been imported we can analyze them using the tools Event Log Explorer provides.



As our analysis shows, the *Brute Force.evtx* log file logs several logon failures before hitting a success, implying that a brute force attack was initiated and successfully executed on the Windows Server 2022 target. The *Event_logs.evtx* log file from the target machine confirms the successful login post compromise.

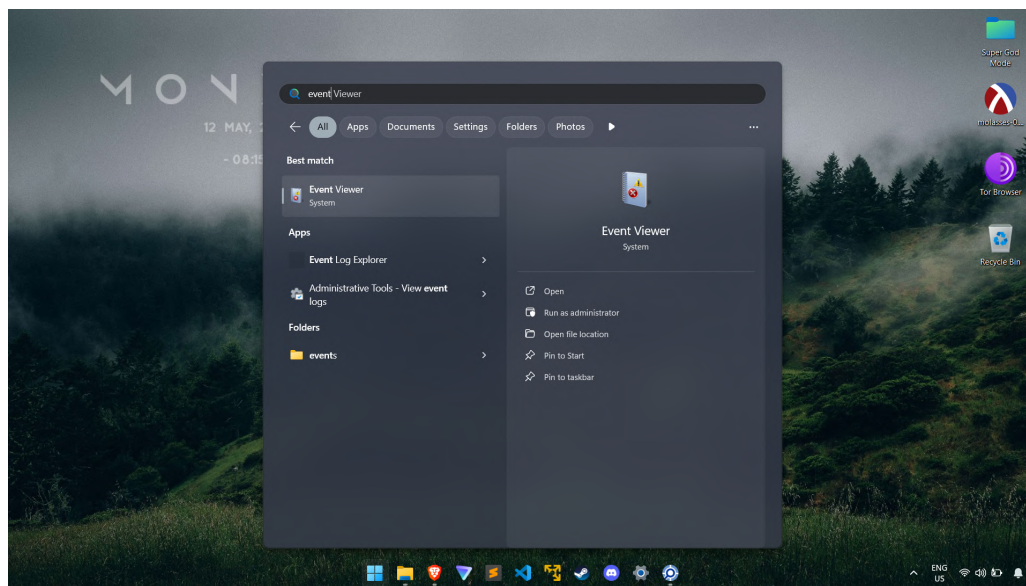
Type	Date	Time	Event	Source	Category	User	Computer
Audit Success	28-10-2020	17:12:22	4672	Microsoft-Windows-Special Logon	N/A	N/A	WIN-SPVSTENBUR3
Audit Success	28-10-2020	17:12:22	4624	Microsoft-Windows Logon	N/A	N/A	WIN-SPVSTENBUR3
Audit Success	28-10-2020	17:12:22	4648	Microsoft-Windows Logon	N/A	N/A	WIN-SPVSTENBUR3
Audit Success	28-10-2020	17:12:22	4776	Microsoft-Windows Credential Validation	N/A	N/A	WIN-SPVSTENBUR3
Audit Success	28-10-2020	17:12:15	4798	Microsoft-Windows User Account Manager	N/A	N/A	WIN-SPVSTENBUR3
Audit Success	28-10-2020	17:12:14	4672	Microsoft-Windows Special Logon	N/A	N/A	WIN-SPVSTENBUR3
Audit Success	28-10-2020	17:12:14	4672	Microsoft-Windows Special Logon	N/A	N/A	WIN-SPVSTENBUR3
Audit Success	28-10-2020	17:12:14	4624	Microsoft-Windows Logon	N/A	N/A	WIN-SPVSTENBUR3
Audit Success	28-10-2020	17:12:14	4624	Microsoft-Windows Logon	N/A	N/A	WIN-SPVSTENBUR3
Audit Success	28-10-2020	17:12:14	4648	Microsoft-Windows Logon	N/A	N/A	WIN-SPVSTENBUR3
Audit Success	28-10-2020	17:09:28	4672	Microsoft-Windows Special Logon	N/A	N/A	WIN-SPVSTENBUR3
Audit Success	28-10-2020	17:09:28	4624	Microsoft-Windows Logon	N/A	N/A	WIN-SPVSTENBUR3
Audit Success	28-10-2020	17:09:28	4648	Microsoft-Windows Logon	N/A	N/A	WIN-SPVSTENBUR3
Audit Success	28-10-2020	17:09:28	4776	Microsoft-Windows Credential Validation	N/A	N/A	WIN-SPVSTENBUR3
Audit Failure	28-10-2020	17:09:28	4625	Microsoft-Windows Logon	N/A	N/A	WIN-SPVSTENBUR3
Audit Failure	28-10-2020	17:09:28	4625	Microsoft-Windows Logon	N/A	N/A	WIN-SPVSTENBUR3
Audit Failure	28-10-2020	17:09:28	4625	Microsoft-Windows Logon	N/A	N/A	WIN-SPVSTENBUR3
Audit Failure	28-10-2020	17:09:28	4625	Microsoft-Windows Logon	N/A	N/A	WIN-SPVSTENBUR3
Audit Failure	28-10-2020	17:09:28	4625	Microsoft-Windows Logon	N/A	N/A	WIN-SPVSTENBUR3
Audit Failure	28-10-2020	17:09:28	4625	Microsoft-Windows Logon	N/A	N/A	WIN-SPVSTENBUR3
Audit Failure	28-10-2020	17:09:28	4625	Microsoft-Windows Logon	N/A	N/A	WIN-SPVSTENBUR3
Audit Success	28-10-2020	16:13:51	4799	Microsoft-Windows Security Group Manager	N/A	N/A	WIN-SPVSTENBUR3
Audit Success	28-10-2020	16:13:51	4799	Microsoft-Windows Security Group Manager	N/A	N/A	WIN-SPVSTENBUR3
Audit Failure	28-10-2020	16:13:51	4625	Microsoft-Windows Logon	N/A	N/A	WIN-SPVSTENBUR3
Audit Failure	28-10-2020	16:13:51	4625	Microsoft-Windows Logon	N/A	N/A	WIN-SPVSTENBUR3
Audit Success	28-10-2020	15:59:39	4798	Microsoft-Windows User Account Manager	N/A	N/A	WIN-SPVSTENBUR3
Audit Success	28-10-2020	15:59:39	4798	Microsoft-Windows User Account Manager	N/A	N/A	WIN-SPVSTENBUR3
Audit Success	28-10-2020	15:59:39	4798	Microsoft-Windows User Account Manager	N/A	N/A	WIN-SPVSTENBUR3

Description
A security-enabled local group membership was enumerated.

Subject:
Description | Data

3.2 Methodology: Event Viewer

Windows also provides an inbuilt tool for viewing and analyzing event logs called Event Viewer. Once we live boot from a Windows Disk Image, we can search for *event* in the **Start Menu** to open the Event Viewer.



Event Viewer allows us to view and analyze event log entries in our disk image.

